

N+ - Network Fundamentals and building network

❖ Network Cable and connectors

1. Draw a labelled diagram of a Cat6 Ethernet cable, showing the color order of wires in the T568B standard and labelling the RJ-45 connector pins.

ANS:

➤ T568B Color Order (RJ-45 Connector)

Pin Number	Wire Color
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown

➤ Simple RJ-45 Connector Diagram

- RJ-45 Connector (Front View)

 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |

| WO | O | WG | B | WB | G | WBr | Br |

2. List three types of network cables and for each, mention one real-world app or device where it would be used, explaining why.

ANS:

Cable Type	Real-World Device/Application	Reason
Cat5e	WIFI Router	Provides reliable internet connectivity for homes and small offices.
USB Cable	Printer	Connects a printer directly to a computer.
Fiber Optic Cable	IPL Streaming Server	Handles very high bandwidth and long-distance data transmission with minimal signal loss.

3. Watch a YouTube video showing how to crimp an RJ-45 connector onto a network cable, then write down the step-by-step process in your own words and mention two common mistakes to avoid.

ANS:

➤ Steps to Crimp an RJ-45 Connector

- Cut the network cable to the required length.
- Strip about 1 inch of the outer jacket using a cable stripper.
- Untwist the wire pairs and arrange them according to the T568B color standard.
- Straighten the wires and trim them evenly.
- Insert the wires into the RJ-45 connector, ensuring each wire reaches the end.
- Verify the color order is correct.
- Place the connector into the crimping tool.
- Squeeze the crimping tool firmly to secure the connector.
- Test the cable using a network cable tester.

➤ Two Common Mistakes to Avoid

- Using the wrong wire color order (not following T568B).
- Not pushing the wires fully into the RJ-45 connector before crimping.

4. Given the following scenario: You are setting up a gaming zone for an IPL screening event and need to connect multiple screens and consoles. Choose the appropriate cable and connector types for high-speed, reliable connections, and explain your choices.

ANS:

➤ Recommended Cable and Connector

- Cable Type: Cat6 Ethernet Cable
- Connector Type: RJ-45 Connector

➤ Explanation

Cat6 cables provide high-speed data transmission up to 1 Gbps or more and reduce interference. RJ-45 connectors are the standard connectors used with Ethernet cables. This combination ensures reliable and stable connectivity for gaming consoles, streaming devices, and display systems during the IPL screening event.

5. Find and share a photo (from your home, college, or online) of any network cable or connector in use and describe what type it is and what devices it connects.

ANS:

➤ Example Description

- I observed a grey Ethernet cable connected to a WiFi router at home.

➤ Cable Type: Cat6 Ethernet Cable

- Connector Type: RJ-45
- Connected Devices: WIFI Router and Desktop Computer

➤ Purpose

- The cable provides a wired internet connection between the router and computer, offering faster speeds and more reliable connectivity than WiFi.

➤ Work

- Sharing a photo from my home, College or online.